

Enter hash table size: 10
Enter number of keys: 7

Enter 7 keys:

23
43
13
27
33
53
63

1 - Linear Probing
2 - Quadratic Probing
3 - Double Hashing
4 - Exit

Enter your choice: 1

Key 23 inserted at index 3
Locations calculated: 1
Key 43 inserted at index 4
Locations calculated: 2
Key 13 inserted at index 5
Locations calculated: 3
Key 27 inserted at index 7
Locations calculated: 1
Key 33 inserted at index 6
Locations calculated: 4
Key 53 inserted at index 8
Locations calculated: 6
Key 63 inserted at index 9
Locations calculated: 7

Hash Table:

Index 0 : EMPTY
Index 1 : EMPTY
Index 2 : EMPTY
Index 3 : 23
Index 4 : 43
Index 5 : 13
Index 6 : 33
Index 7 : 27
Index 8 : 53
Index 9 : 63

Total locations calculated = 24

Total locations calculated = 24
Number of keys inserted = 7
Average locations calculated = 3.43
1 - Linear Probing
2 - Quadratic Probing
3 - Double Hashing
4 - Exit

Enter your choice: 2

Key 23 inserted at index 3
Locations calculated: 1
Key 43 inserted at index 4
Locations calculated: 2
Key 13 inserted at index 7
Locations calculated: 3
Key 27 inserted at index 8
Locations calculated: 2
Key 33 inserted at index 2
Locations calculated: 4
Key 53 inserted at index 9
Locations calculated: 5
Hash table is full!

Hash Table:

Index 0 : EMPTY
Index 1 : EMPTY
Index 2 : 33
Index 3 : 23
Index 4 : 43
Index 5 : EMPTY
Index 6 : EMPTY
Index 7 : 13
Index 8 : 27
Index 9 : 53

Total locations calculated = 17
Number of keys inserted = 7
Average locations calculated = 2.43
1 - Linear Probing
2 - Quadratic Probing
3 - Double Hashing
4 - Exit

Enter your choice: 3

Key 23 inserted at index 3
Locations calculated: 1

Number of keys inserted = 7
Average locations calculated = 2.43

1 - Linear Probing
2 - Quadratic Probing
3 - Double Hashing
4 - Exit

Enter your choice: 3

Key 23 inserted at index 3
Locations calculated: 1
Key 43 inserted at index 1
Locations calculated: 2
Key 13 inserted at index 8
Locations calculated: 2
Key 27 inserted at index 7
Locations calculated: 1
Key 33 inserted at index 0
Locations calculated: 2
Key 53 inserted at index 2
Locations calculated: 2
Key 63 inserted at index 4
Locations calculated: 2

Hash Table:

Index 0 : 33
Index 1 : 43
Index 2 : 53
Index 3 : 23
Index 4 : 63
Index 5 : EMPTY
Index 6 : EMPTY
Index 7 : 27
Index 8 : 13
Index 9 : EMPTY

Total locations calculated = 12
Number of keys inserted = 7
Average locations calculated = 1.71

1 - Linear Probing
2 - Quadratic Probing
3 - Double Hashing
4 - Exit

Enter your choice: 4

Program terminated.

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